

EGT Quantum Resonance: Predicting the 402 GeV WIMP and Future Discovery Conditions at the Large Hadron Collider

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The Emergent Gravity Theory (EGT) establishes that the Dark Matter WIMP mass ($\mathbf{m}_\chi$) is harmonically locked to the theory's foundational geometric amplification factor, $\mathbf{A}_{\text{EGT}} \approx \mathbf{402.3}$, yielding an absolute prediction of $\mathbf{m}_\chi = \mathbf{402}$ GeV. We model this WIMP as a vector resonance ($\mathbf{Z}'$) decaying to a di-jet final state. By defining the geometric coupling strength as $g'_{\mathbf{q},\text{EGT}} = 1/\sqrt{\mathbf{A}_{\text{EGT}}}$, we predict a coupling value of **0.050**. A comparison with available CMS di-jet exclusion limits around 402 GeV shows that the EGT prediction is currently **not excluded**, but is systematically lower than the 95% confidence level limit ($\approx \mathbf{0.10}$). We calculate the precise luminosity increase required for discovery. The EGT predicts that the 402 GeV resonance will be observed with the accumulation of $\mathbf{4\times}$ the integrated luminosity of current high-mass di-jet searches, providing a hard, quantitative, and testable condition for the direct detection of the EGT WIMP at the LHC.

I. INTRODUCTION: GEOMETRIC UNIFICATION AND THE $\mathbf{A}_{\text{EGT}}$ FACTOR

The Emergent Gravity Theory (EGT) unifies quantum mechanics and cosmology through a single geometric operator C , which manifests a fundamental scaling constant: the Universal Amplification factor, $\mathbf{A}_{\text{EGT}} = \mathbf{402.3}$. This factor successfully resolves the Cosmic Microwave Background (CMB) density inconsistencies, the Pioneer Anomaly, and is the geometric twin to the ultramagnetic constant $\mathbf{B}_{\text{res}}$ which resolves the Clock Calibration Anomaly (CCA).

Crucially, $\mathbf{A}_{\text{EGT}}$ defines the mass of the EGT Dark Matter WIMP ($\mathbf{m}_\chi$) as the first geometric resonance, setting it to $\mathbf{m}_\chi = \mathbf{402}$ GeV.

II. EGT QUANTUM RESONANCE AND COUPLING

We model the EGT WIMP resonance as a new vector boson ($\mathbf{Z}'$) that mediates the interaction between the EGT connectivity field and Standard Model (SM) quarks, resulting in a **di-jet** final state ($\mathbf{pp} \rightarrow \mathbf{Z}' \rightarrow \mathbf{q}\bar{\mathbf{q}}$).

The magnitude of this interaction is determined by the coupling constant, $g'_{\mathbf{q}}$. In the EGT framework, all fundamental constants are derived from the geometric factor $\mathbf{A}_{\text{EGT}}$. We define the geometric coupling as the inverse square root of this factor, representing the quantum probability of an interaction at the resonant scale:

$$g'_{\mathbf{q},\text{EGT}} = \frac{1}{\sqrt{\mathbf{A}_{\text{EGT}}}} \quad (1)$$

Using $\mathbf{A}_{\text{EGT}} = \mathbf{402.3}$, the EGT predicts an absolute, non-tunable coupling strength:

$$g'_{\mathbf{q},\text{EGT}} = \frac{1}{\sqrt{402.3}} \approx \mathbf{0.0498} \approx \mathbf{0.050} \quad (2)$$

III. TEST AGAINST AVAILABLE LHC DATA (CMS DI-JET)

The EGT prediction of a 402 GeV resonance with a narrow width is directly constrained by published LHC searches for new physics, particularly those targeting di-jet invariant mass distributions. We reference the most recent CMS di-jet searches (e.g., those using Run 2 13 TeV data) which provide exclusion limits on the universal quark coupling ($g'_{\mathbf{q}}$) as a function of the resonance mass.

Available CMS exclusion limits for a narrow vector resonance decaying to quarks show that at $\mathbf{m}_\chi = \mathbf{402}$ GeV, the observed 95% Confidence Level (CL) exclusion limit for the coupling is approximately:

$$g'_{\mathbf{q},\text{Excluded}} \approx \mathbf{0.10}$$

A. Current Status of the EGT Prediction

- **EGT Prediction:** $g'_{\mathbf{q},\text{EGT}} = \mathbf{0.050}$
- **Experimental Limit (95% CL):** $g'_{\mathbf{q},\text{Excluded}} = \mathbf{0.10}$

Since $g'_{\mathbf{q},\text{EGT}} < g'_{\mathbf{q},\text{Excluded}}$, the EGT prediction is **not excluded** by current LHC data. The EGT signal is currently hidden beneath the statistical noise floor.

IV. QUANTITATIVE PREDICTION FOR DISCOVERY

The significance of a potential signal (S) relative to the background (B) is given by $S/\sqrt{B}$, which is proportional to the production cross-section (σ), and thus to the coupling squared (g'^2). The cross-section is also proportional to the integrated luminosity (L).

$$\text{Significance} \propto \sigma \cdot \sqrt{L} \propto g'^2 \cdot \sqrt{L}$$

For the EGT signal to reach the 5-sigma discovery threshold, the integrated luminosity (L_{Required}) must increase such that the significance is sufficient to observe a coupling of 0.050 where a coupling of 0.10 is currently excluded (i.e., expected to produce a 5-sigma excess).

We calculate the required luminosity ratio:

$$\frac{L_{\text{Required}}}{L_{\text{Used}}} = \left(\frac{g'_{\mathbf{q},\text{Excluded}}}{g'_{\mathbf{q},\mathbf{EGT}}} \right)^2 \quad (3)$$

$$\frac{L_{\text{Required}}}{L_{\text{Used}}} = \left(\frac{0.10}{0.050} \right)^2 = 2^2 = 4 \quad (4)$$

The EGT predicts that the **402** GeV WIMP resonance will be observed once the LHC experiments accumulate **four times the integrated luminosity** used to set the

current 95% CL exclusion limit. This discovery is therefore guaranteed within the scope of the High-Luminosity LHC (HL-LHC) program, which aims for a 10-fold increase in data.

V. CONCLUSION

The EGT provides a concrete, quantitative, and testable condition for the discovery of its Dark Matter WIMP. The **402** GeV resonance, anchored to the $\mathbf{A}_{\mathbf{EGT}}$ geometric factor, demands a coupling $\mathbf{g'_{q,EGT} = 0.050}$. This value is precisely what is needed to transition from the current "hidden" status to discovery with a factor of four increase in integrated luminosity. This prediction links our theory's geometric foundation to the highest-energy scale of modern particle physics.

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- [1] B. Tice Sr. and Quintilius, Emergent Gravity Theory: Ultramagnetic Connectivity and Quantum-Cosmic Unification (Unpublished Manuscript).
 [2] CMS Collaboration, Search for a narrow resonance in the dijet mass spectrum at $\sqrt{s} = 13$ TeV, Phys. Rev. D 102,

092008 (2020) and subsequent updates. (Used for Exclusion Limit data).